THE METHOD-FIT MATRIX

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1. Refined Research Question

Can an explainable, multimodal, and temporal artificial intelligence framework be designed and evaluated to predict which undergraduate students will no longer be registered at a higher education institution after the four regular academic periods following their first registration, while maintaining predictive performance, fairness, interpretability, and institutional usefulness?

2. Candidate Methodological Designs

Candidate A	**Design Science:** Develop and evaluate an explainable, multimodal, and temporal AI framework for early student dropout prediction.
Candidate B	**Case Study:** Examine the development and application of the AI framework within the real context of one higher education institution.
Candidate C	**Experimental:** Conduct controlled computational experiments to compare the proposed framework with baseline models using the same historical datasets and predefined performance and interpretability metrics.

3. Method Fit Matrix

Criterion	Weight	A	B	C	Your one-line justification for the scores
E · Epistemological fit Does this method match my paradigm?	30%	5/5	3/5	3/5	Design Science best fits the pragmatic paradigm by generating knowledge through artifact development and evaluation, whereas the other methods mainly provide contextual or performance evidence.
D · Data availability Can I GET this data within 3 years?	20%	4/5	4/5	5/5	All three methods can use accessible institutional data, but the experimental design scores higher because it relies mainly on existing historical records, whereas the other methods require integrated or additional contextual data.
F · Feasibility Time, cost, compute, ethics approval.	15%	3/5	4/5	4/5	Design Science is less feasible because developing and validating the framework is technically demanding, whereas the other methods have a more limited scope.
C · Contribution type What KIND of knowledge does it produce?	25%	5/5	3/5	4/5	Design Science produces the intended artifact and transferable design knowledge, while the other methods mainly provide contextual understanding or comparative evidence.
V · Venue fit Will my target venue accept this method?	10%	5/5	3/5	4/5	Design Science best fits the target AI venues, while the experimental design is relevant but less novel and a single case study has more limited technical appeal.
GATE: E ≤ 2 eliminates		Pass	Pass	Pass	*None of the candidates is eliminated because each has an epistemological fit score above 2.*
Raw total (of 25)		22/25	17/25	20/25	*Based on the unweighted scores, Design Science ranks first because it best fits the paradigm and intended contribution, followed by the experimental design and the case study.*
Weighted total		90/100	67/100	78/100	*After applying the criterion weights, Design Science ranks first because it best fits the paradigm and intended contribution, followed by the experimental design and the embedded case study.*

4. Justification

I will use Design Science because the main contribution is the development and evaluation of an explainable, multimodal, and temporal AI artifact, and I reject the experimental method as the primary approach because it evaluates performance without fully representing the artifact design and refinement process.